

Lectures 9–11: Gaussian Processes

Feature maps, maxima, and the geometry of the index set

OUTLINE L9 Definition $\rightarrow$ feature maps: a kernel is an inner product $\rightarrow$ examples: Mu on the sphere, Brownian motion, the wide network $\rightarrow$ maxima: $\sqrt{2 \log n}$, with or without independence

L10 Suprema: independent Gaussians reach $\sqrt{2 \log n} \rightarrow$ covering numbers $\rightarrow$ Dudley's inequality: statement, the sphere, proof by chaining $\rightarrow$ Sudakov's lower bound, stated

L11 (*planned*) Comparison: Sudakov–Fernique and Slepian by Gaussian interpolation $\rightarrow$ the operator norm, $\mathbb{E}\|M\| \leq \sqrt{N} + \sqrt{d} \rightarrow$ Sudakov's lower bound, proved $\rightarrow$ what the midterm covers

HOW TO READ Two facts are quoted in boxes, one per lecture. Exercises are typed A (did you follow the lecture), B (did you follow the argument) and C (a small theory to build yourself).

1 Gaussian processes

Q. What is a Gaussian process, and what is the first thing to ask about one?

A Gaussian vector indexed by an arbitrary set T . Its covariance $\Sigma_{s,t}$ is an inner product $\langle \phi(s), \phi(t) \rangle$ of *features*, so the process is the set $\{\phi(t)\}$ seen through one standard Gaussian. The basic quantity is $\mathbb{E} \sup_t X_t$.

Two Gaussian processes have already appeared: the output of a wide random network at finitely many inputs (Lecture 3), a Gaussian vector with covariance $\Sigma_{x,x'} = \mathbb{E}[\sigma(\langle w, x \rangle) \sigma(\langle w, x' \rangle)]$; and $u \mapsto Mu$ for a Gaussian matrix M (Lecture 5), where $\mathbb{E}\|Mu\|^2$ was computed for fixed u . Lectures 9–11 bound $\sup_u \|Mu\|$, the operator norm, and more generally the maximum of a Gaussian process over its index set.

1.1 The definition

Definition 1.1 (Gaussian process). Let T be any set. A family of real random variables $X = \{X_t\}_{t \in T}$ is a *Gaussian process* if for every finite $t_1, \dots, t_n \in T$ the vector $(X_{t_1}, \dots, X_{t_n})$ is a Gaussian vector in the sense of Definition 1.3 of Lectures 1–2; equivalently, by Corollary 1.7 there, if every finite linear combination $\sum_j a_j X_{t_j}$ is a one-dimensional Gaussian. Its *mean function* and *covariance kernel* are

$$\mu_t = \mathbb{E}X_t, \quad \Sigma_{s,t} = \text{Cov}(X_s, X_t).$$

The process is *centered* if $\mu \equiv 0$. All processes below are centered unless said otherwise.

Every finite marginal of X is determined by μ and Σ , and nothing else about X will be used.

Lemma 1.2 (Kernels are positive semidefinite). *For every $t_1, \dots, t_n \in T$ and $a \in \mathbb{R}^n$,*

$$\sum_{j,k} a_j a_k \Sigma_{t_j, t_k} \geq 0.$$

Proof. It is $\text{Var}(\sum_j a_j X_{t_j})$. ■

The converse — every positive semidefinite Σ is the kernel of a Gaussian process — is stated in §1.2.

1.2 Feature maps: a kernel is an inner product

If $\phi : T \rightarrow \mathbb{R}^k$ is any map and $g \sim \mathcal{N}(0, I_k)$, then

$$X_t = \langle \phi(t), g \rangle \quad (1)$$

is a Gaussian process: any finite collection $(X_{t_1}, \dots, X_{t_n})$ is the linear image Ag of g , with rows $\phi(t_j)$, so it is Gaussian by Theorem 1.5 of Lectures 1–2, with covariance $AA^\top$, that is

$$\Sigma_{s,t} = \langle \phi(s), \phi(t) \rangle.$$

The map ϕ is a *feature map* for Σ .

In some examples $\phi(t)$ is a function rather than a vector, and H is a *Hilbert space*: a space with an inner product in which limits can be taken. Three occur: $\mathbb{R}^k$; $L^2[0, 1]$, the square-integrable functions on $[0, 1]$ with $\langle f, h \rangle = \int_0^1 fh$; and L^2 of a Gaussian, the functions of a Gaussian vector w with $\langle f, h \rangle = \mathbb{E}[f(w)h(w)]$. Each is $\mathbb{R}^k$ with k allowed to be infinite: for an orthonormal basis $e_1, e_2, \dots$ of H and i.i.d. standard Gaussians $g_1, g_2, \dots$, (1) reads $X_t = \sum_i \langle \phi(t), e_i \rangle g_i$, which converges since $\sum_i \langle \phi(t), e_i \rangle^2 = \|\phi(t)\|^2 < \infty$. We write $X_t = \langle \phi(t), g \rangle$ in every case.

Quoted: feature maps produce processes, and every kernel has one. (i) If $\phi : T \rightarrow H$ is any map into a Hilbert space, then $X_t = \langle \phi(t), g \rangle$ is a centered Gaussian process with kernel $\Sigma_{s,t} = \langle \phi(s), \phi(t) \rangle$. (ii) Conversely, every symmetric positive semidefinite Σ on T is of this form for some Hilbert space H and some $\phi : T \rightarrow H$. Part (i) for $H = \mathbb{R}^k$ is the argument above; part (ii) is the Moore–Aronszajn theorem. Every kernel below comes with an explicit feature map, so (ii) is not used.

Remark 1.3 (What the process remembers). ϕ and $U\phi$, with U a rotation of H , give the same kernel and so the same process. The process depends on ϕ only through the geometry of the feature set $\{\phi(t) : t \in T\}$: lengths $\|\phi(t)\|^2 = \Sigma_{t,t}$, distances, angles, convex hull. The maximum (§1.5) and the zero set (Exercise 1.16) are two functions of that geometry.

1.3 Examples

Example 1.4 (Finite index set). $T = \{1, \dots, n\}$ and Σ an $n \times n$ covariance matrix. Write $\Sigma = AA^\top$ and let $\phi(j)$ be the j th row of A . Then $X_j = \langle \phi(j), g \rangle = (Ag)_j$: the feature map of a Gaussian vector is a square root of its covariance.

Example 1.5 (The sphere, through a Gaussian matrix). Let M be an $N \times d$ matrix with i.i.d. $\mathcal{N}(0, 1)$ entries and let $T = S^{d-1}$, the unit sphere in $\mathbb{R}^d$. For $u \in T$ put $X_u = Mu \in \mathbb{R}^N$. Each coordinate is $(Mu)_i = \langle M_i, u \rangle$, with M_i the i th row, so for every u, v and i, j ,

$$\mathbb{E}[(Mu)_i (Mv)_j] = \sum_{k,l} u_k v_l \mathbb{E}[M_{ik} M_{jl}] = \delta_{ij} \langle u, v \rangle.$$

The N coordinate processes are independent copies of one scalar process on the sphere with kernel

$$\Sigma_{u,v} = \langle u, v \rangle,$$

whose feature map is the identity, $\phi(u) = u \in \mathbb{R}^d$: the process is $u \mapsto \langle g, u \rangle$ for a single $g \sim \mathcal{N}(0, I_d)$, and the feature set is the sphere itself. Its maximum is explicit,

$$\sup_{u \in S^{d-1}} \langle g, u \rangle = \|g\|,$$

attained at $u = g/\|g\|$. The whole matrix is the process $Y_{(u,v)} = \langle v, Mu \rangle = \sum_{ij} v_i M_{ij} u_j$ indexed by $S^{N-1} \times S^{d-1}$, with kernel $\langle v, v' \rangle \langle u, u' \rangle$ and feature map $\phi(u, v) = vu^\top$, a rank-one matrix of Frobenius norm one; its maximum is the operator norm,

$$\sup_{u,v} \langle v, Mu \rangle = \|M\|_{\text{op}} = \sup_u \|Mu\|.$$

For fixed unit u , $\mathbb{E}\|Mu\|^2 = N$; Lecture 11 shows $\mathbb{E}\|M\|_{\text{op}} \leq \sqrt{N} + \sqrt{d}$.

Example 1.6 (Brownian motion). $T = [0, 1]$, $H = L^2[0, 1]$, and $\phi(t) = \mathbf{1}_{[0,t]}$, the indicator function of $[0, t]$. Then

$$\Sigma_{s,t} = \int_0^1 \mathbf{1}_{[0,s]} \mathbf{1}_{[0,t]} = \min(s, t),$$

and the process B_t with this kernel is *Brownian motion*: $B_0 = 0$, $B_t \sim \mathcal{N}(0, t)$, and increments over disjoint intervals are independent, because $\mathbf{1}_{(s,t]}$ and $\mathbf{1}_{[0,r]}$ are orthogonal for $r \leq s$ (Exercise 1.12). The features are indicator functions, and the distance between two of them is $\|\mathbf{1}_{[0,t]} - \mathbf{1}_{[0,s]}\| = \sqrt{|t - s|}$.

Example 1.7 (The wide network). Lecture 3, Proposition 1.24: for inputs $x \in \mathbb{R}^d$, weights $w \sim \mathcal{N}(0, I_d)$ and an activation σ with $\mathbb{E}\sigma(\langle w, x \rangle)^2 < \infty$, the outputs of a wide random network at finitely many inputs converge to a Gaussian vector with kernel $\Sigma_{x,x'} = \mathbb{E}[\sigma(\langle w, x \rangle)\sigma(\langle w, x' \rangle)]$. The feature map is

$$\phi(x) = \sigma(\langle \cdot, x \rangle) \in H = L^2(\mathcal{N}(0, I_d)),$$

the activation at x viewed as a function of the weight w ; then $\langle \phi(x), \phi(x') \rangle = \mathbb{E}[\sigma(\langle w, x \rangle)\sigma(\langle w, x' \rangle)]$ is exactly $\Sigma_{x,x'}$. The limit of Lecture 3 is the Gaussian process with this feature map. Conditioning it on data is Theorem 2.9 of Lectures 1–2 with the matrix $(\Sigma_{x_i, x_j})_{ij}$: an infinitely wide network under exact Bayesian inference is kernel regression with this Σ (Exercise 1.13 does the computation for Brownian motion).

1.4 The canonical metric

Definition 1.8. The *canonical metric* of a centered Gaussian process is

$$d(s, t) = \|X_s - X_t\|_{L^2} = (\mathbb{E}(X_s - X_t)^2)^{1/2} = \sqrt{\Sigma_{s,s} - 2\Sigma_{s,t} + \Sigma_{t,t}}.$$

Through a feature map, $d(s, t) = \|\phi(s) - \phi(t)\|$. On the sphere (Example 1.5) it is the Euclidean distance $\|u - v\|$; for Brownian motion it is $\sqrt{|t - s|}$. Lecture 10 bounds the maximum of a process in terms of this metric.

1.5 Maxima

Fix a centered Gaussian process on T . For finite T the maximum $\max_t X_t$ is an ordinary random variable; for general T we define

$$\mathbb{E} \sup_{t \in T} X_t := \sup \left\{ \mathbb{E} \max_{t \in F} X_t : F \subset T \text{ finite} \right\}, \quad (2)$$

which may be $+\infty$ and involves only finite marginals.

Proposition 1.9 (The union bound). Let $X_1, \dots, X_n$ be centered Gaussians, not necessarily independent, with $\text{Var } X_i \leq \sigma^2$. Then

$$\mathbb{E} \max_{i \leq n} X_i \leq \sigma \sqrt{2 \log n}.$$

Proof. For $\lambda > 0$, by Jensen and then the union bound on the maximum of positive numbers,

$$e^{\lambda \mathbb{E} \max_i X_i} \leq \mathbb{E} e^{\lambda \max_i X_i} = \mathbb{E} \max_i e^{\lambda X_i} \leq \sum_i \mathbb{E} e^{\lambda X_i} = \sum_i e^{\lambda^2 \text{Var } X_i / 2} \leq n e^{\lambda^2 \sigma^2 / 2}.$$

Take logarithms: $\mathbb{E} \max_i X_i \leq \frac{\log n}{\lambda} + \frac{\lambda \sigma^2}{2}$, and choose $\lambda = \sqrt{2 \log n} / \sigma$. ■

The bound does not use independence. It is sharp for independent X_i (Proposition 2.1), and correlation lowers the maximum (Exercise 1.15(a)). On the sphere, Example 1.5, $\mathbb{E} \sup_u \langle g, u \rangle = \mathbb{E} \|g\| \in [\sqrt{d} - 1, \sqrt{d}]$ (Exercise 1.10), the value the union bound gives for $n = e^{d/2}$ points. Through a feature map, $\sup_t X_t = \sup_{h \in S} \langle h, g \rangle$ with S the feature set; its expectation is the *Gaussian width* of S (Exercise 1.14).

1.6 What comes next

- (i) **Chaining** (Lecture 10). If $N(\varepsilon)$ balls of radius ε in the canonical metric cover T , then $\mathbb{E} \sup X \leq 12 \int_0^\infty \sqrt{\log N(\varepsilon)} d\varepsilon$ (Dudley). On the sphere this is $\sqrt{d}$ up to a constant. Sudakov's inequality is the lower bound in terms of $N(\varepsilon)$.
- (ii) **Comparison** (Lecture 11). If $d_X(s, t) \leq d_Y(s, t)$ for all s, t then $\mathbb{E} \sup X \leq \mathbb{E} \sup Y$ (Sudakov–Fernique), by Gaussian interpolation and integration by parts (Lecture 5). It gives Sudakov's inequality and $\mathbb{E} \|M\|_{\text{op}} \leq \sqrt{N} + \sqrt{d}$.

Exercise 1.10 (TYPE A). Let M be $N \times d$ with i.i.d. $\mathcal{N}(0, 1)$ entries. (a) Verify $\mathbb{E}[(Mu)_i(Mv)_j] = \delta_{ij} \langle u, v \rangle$, and conclude that for unit vectors u, v the pair Mu, Mv is jointly Gaussian with $\mathbb{E} \langle Mu, Mv \rangle = N \langle u, v \rangle$. (b) For $g \sim \mathcal{N}(0, I_d)$ show that $\sup_{u \in S^{d-1}} \langle g, u \rangle = \|g\|$ and $\mathbb{E} \|g\| \leq \sqrt{d}$. (c) Show $\mathbb{E} \|g\| \geq \sqrt{d} - 1$. Compute $\mathbb{E} \|g\|^4 = d^2 + 2d$ and use $\mathbb{E} Y^2 \leq (\mathbb{E} Y)^{2/3} (\mathbb{E} Y^4)^{1/3}$. For $d = 100$ the truth is 9.975.

Exercise 1.11 (TYPE A). Kernels combine. Let Σ_1, Σ_2 be kernels on T with feature maps ϕ_1, ϕ_2 into $\mathbb{R}^{k_1}, \mathbb{R}^{k_2}$. (a) Write down feature maps for $\Sigma_1 + \Sigma_2$ and for the product $\Sigma_1 \Sigma_2$, so both are kernels. (b) Deduce that $\langle u, v \rangle^m$ is a kernel on $\mathbb{R}^d$ for every m . (c) Write down the Gaussian process with kernel $\langle u, v \rangle^2$ explicitly: it is $u \mapsto u^\top G u$ for a $d \times d$ matrix G with i.i.d. Gaussian entries. In (a), stack the two feature vectors for the sum, and take the $k_1 k_2$ products $\phi_1(t)_a \phi_2(t)_b$ for the product.

Exercise 1.12 (TYPE A). Brownian motion has $\Sigma_{s,t} = \min(s, t)$ on $[0, 1]$. (a) Check $\langle \mathbf{1}_{[0,s]}, \mathbf{1}_{[0,t]} \rangle = \min(s, t)$ in $L^2[0, 1]$. (b) Show that for $0 \leq r \leq s \leq t \leq 1$ the increment $B_t - B_s$ is $\mathcal{N}(0, t - s)$ and independent of B_r . Compute one covariance, then use Theorem 1.16 of Lectures 1–2.

Exercise 1.13 (TYPE B). Conditioning a Gaussian process on finitely many of its values is Theorem 2.9 of Lectures 1–2 with the matrix $(\Sigma_{t_i, t_j})_{ij}$. (a) For Brownian motion compute $\mathbb{E}[B_t | B_1]$ and $\text{Var}(B_t | B_1)$ for $t \in [0, 1]$. (b) Condition instead on $B_{t_1}, \dots, B_{t_n}$ with $0 < t_1 < \dots < t_n < 1$, and show that $t \mapsto \mathbb{E}[B_t | B_{t_1}, \dots, B_{t_n}]$ is the piecewise linear interpolation of the observed values. Guess the answer, and check that $B_t - (\text{guess})$ is uncorrelated with every B_{t_j} ; independent increments do the work. The process $B_t - tB_1$ in (a) is the Brownian bridge.

Exercise 1.14 (TYPE C). *The Gaussian width.* For $S \subset \mathbb{R}^k$ define

$$w(S) = \mathbb{E} \sup_{h \in S} \langle h, g \rangle, \quad g \sim \mathcal{N}(0, I_k).$$

(a) Explain why, for a process with feature map $\phi : T \rightarrow \mathbb{R}^k$, $\mathbb{E} \sup_{t \in T} X_t = w(S)$ with $S = \{\phi(t) : t \in T\}$: the expected maximum of a process is the width of its feature set. (b) Prove the rules: $w(S) \geq 0$; $S \subset S' \Rightarrow w(S) \leq w(S')$; $w(S + x) = w(S)$ and $w(\lambda S) = \lambda w(S)$ for $\lambda \geq 0$; $w(\text{conv } S) = w(S)$. (c) Compute three widths. Write B_2^k, B_1^k, B_∞^k for the unit balls of the Euclidean, ℓ^1 and ℓ^∞ norms, and show

$$w(B_2^k) = \mathbb{E} \|g\|_2 \in [\sqrt{k} - 1, \sqrt{k}], \quad w(B_1^k) = \mathbb{E} \max_i |g_i| \leq \sqrt{2 \log(2k)}, \quad w(B_\infty^k) = \mathbb{E} \|g\|_1 = k \sqrt{\frac{2}{\pi}}.$$

In each case the supremum of $\langle h, g \rangle$ over a unit ball is a norm of g . For B_2^k use Exercise 1.10; for B_1^k apply Proposition 1.9 to the $2k$ Gaussians $\pm g_i$. (d) Let $S_k \subset \mathbb{R}^d$ be the unit vectors with at most k nonzero entries, and show $w(S_k) \leq \sqrt{k} \sqrt{2 \log(2d)}$. To a Gaussian direction, the k -sparse unit vectors have dimension $k \log d$, not d .

Exercise 1.15 (TYPE B). Two ways a maximum can be small. (a) Let $g_0, g_1, \dots, g_n$ be i.i.d. $\mathcal{N}(0, 1)$, fix $\rho \in [0, 1]$, and set $X_i = \sqrt{\rho} g_0 + \sqrt{1 - \rho} g_i$. Compute the kernel of $(X_1, \dots, X_n)$ and show $\mathbb{E} \max_i X_i = \sqrt{1 - \rho} \mathbb{E} \max_i g_i$: correlation shrinks the maximum, and at $\rho = 1$ kills it. (b) Let $u_1, \dots, u_n$ be unit vectors in $\mathbb{R}^d$ and $X_i = \langle u_i, g \rangle$. Show $\mathbb{E} \max_i X_i \leq \min(\sqrt{2 \log n}, \mathbb{E} \|g\|)$. Which configurations of the u_i make the first bound essentially sharp, and which the second? *Two bounds on one maximum: $\sqrt{2 \log n}$ from counting, $\sqrt{d}$ from geometry.*

Exercise 1.16 (TYPE B). The zero set of a process is a random slice of its feature set. Let $\phi : T \rightarrow \mathbb{R}^k$ with $\phi(t) \neq 0$ for all t , $g \sim \mathcal{N}(0, I_k)$, and $X_t = \langle \phi(t), g \rangle$. (a) Show that $\{t : X_t = 0\} = \{t : \phi(t) \in g^\perp\}$, where $g^\perp$ is the hyperplane through the origin normal to g , and that $g^\perp$ is a *uniformly random* hyperplane: its law is invariant under every rotation of $\mathbb{R}^k$. (b) Now let $T = [a, b]$, ϕ smooth, and $\gamma(t) = \phi(t) / \|\phi(t)\|$ the feature curve projected to the unit sphere. Show

$$\mathbb{E} \#\{t \in [a, b] : X_t = 0\} = \frac{1}{\pi} \text{length}(\gamma).$$

By (a) this is the expected number of times γ crosses a random hyperplane. For an arc of a great circle of length ℓ that number is $c\ell$, by rotation invariance and additivity; a full great circle crosses a hyperplane exactly twice, which gives c . A smooth curve is a limit of such arcs. (c) Apply (b) to the random trigonometric polynomial $X_\theta = \sum_{j=1}^n (a_j \cos j\theta + b_j \sin j\theta)$, $\theta \in [0, 2\pi)$, with all a_j, b_j i.i.d. $\mathcal{N}(0, 1)$: identify ϕ , check that $\|\phi(\theta)\|$ is constant, and conclude that the expected number of zeros is $2\sqrt{(n+1)(2n+1)}/6 \approx 2n/\sqrt{3}$. For $n = 10$ a simulation gives 12.44 against 12.41.

ASSIGNED Lecture 9, due at the start of Lecture 10: Exercises 1.10, 1.11, 1.12, 1.13, 1.14, 1.15, 1.16. *Seven problems, most of them short. The first three check the examples; 1.14 builds the one definition the lecture leaves out; 1.16 is a small theorem the lecture does not reach.*

2 Suprema: independence, chaining, and the sphere

Q. How large is $\mathbb{E} \sup_{t \in T} X_t$ when T is infinite?

For n points the union bound gives $\sqrt{2 \log n}$, which is sharp for independent coordinates. For general T , Dudley's inequality: cover T by $N(\varepsilon)$ balls of radius ε in the canonical metric and integrate $\sqrt{\log N(\varepsilon)}$ over ε . On the sphere this gives $\sqrt{d}$ up to a constant; Sudakov's inequality is the matching lower bound.

Throughout, X is a centered Gaussian process on T with canonical metric $d(s, t) = (\mathbb{E}(X_s - X_t)^2)^{1/2}$ (Definition 1.8), and $\mathbb{E} \sup_t X_t$ means (2).

2.1 Independent Gaussians reach $\sqrt{2 \log n}$

Proposition 2.1 (The independent case). *If $g_1, \dots, g_n$ are i.i.d. $\mathcal{N}(0, 1)$ then, as $n \rightarrow \infty$,*

$$\mathbb{E} \max_{i \leq n} g_i = \sqrt{2 \log n} (1 + o(1)).$$

Proof. The upper bound is Proposition 1.9. For the lower bound, first a tail estimate: for $t \geq 1$,

$$\mathbb{P}(g > t) = \frac{1}{\sqrt{2\pi}} \int_t^\infty e^{-x^2/2} dx \geq \frac{1}{\sqrt{2\pi}} \int_t^{t+1/t} e^{-x^2/2} dx \geq \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{t} e^{-(t+1/t)^2/2} \geq \frac{e^{-3/2}}{\sqrt{2\pi} t} e^{-t^2/2},$$

since $(t + 1/t)^2/2 = t^2/2 + 1 + 1/(2t^2) \leq t^2/2 + 3/2$. Fix $\delta \in (0, 1)$ and put $t = \sqrt{(2 - \delta) \log n}$. Then $n \mathbb{P}(g > t) \geq c n^{\delta/2}/t \rightarrow \infty$, so by independence

$$\mathbb{P}\left(\max_i g_i \leq t\right) = (1 - \mathbb{P}(g > t))^n \leq e^{-n \mathbb{P}(g > t)} \rightarrow 0.$$

Finally $\max_i g_i \geq t$ on the event $\{\max_i g_i > t\}$ and $\max_i g_i \geq g_1 \geq -g_1^-$ always, so

$$\mathbb{E} \max_i g_i \geq t \mathbb{P}\left(\max_i g_i > t\right) - \mathbb{E} g_1^- = \sqrt{(2 - \delta) \log n} (1 - o(1)) - \frac{1}{\sqrt{2\pi}}.$$

Divide by $\sqrt{2 \log n}$ and let $n \rightarrow \infty$, then $\delta \downarrow 0$. ■

Reference example: $X_u = \langle g, u \rangle$ on S^{d-1} has $\mathbb{E} \sup_u X_u = \mathbb{E} \|g\| \in [\sqrt{d} - 1, \sqrt{d}]$ (Exercise 1.10), the value of $\sqrt{2 \log n}$ at $n = e^{d/2}$. The canonical metric is $d(u, v) = \|u - v\|$, and the sphere contains $e^{\Theta(d)}$ points at mutual distance $\frac{1}{4}$ (Lemma 2.3, Exercise 2.8).

2.2 Covering numbers

Definition 2.2 (Covering numbers). For $\varepsilon > 0$, an ε -net of (T, d) is a subset $T' \subset T$ such that every $t \in T$ is within distance ε of some point of T' . The *covering number* $N(T, d, \varepsilon)$ is the smallest size of an ε -net ($+\infty$ if there is no finite one). It is nonincreasing in ε , and $N(T, d, \varepsilon) = 1$ once $\varepsilon \geq \text{diam}(T)$.

Lemma 2.3 (The sphere). *For the Euclidean metric on S^{d-1} and every $\varepsilon > 0$,*

$$N(S^{d-1}, \|\cdot\|, \varepsilon) \leq \left(1 + \frac{2}{\varepsilon}\right)^d.$$

Proof. Let $u_1, \dots, u_n \in S^{d-1}$ be ε -separated, $\|u_i - u_j\| > \varepsilon$ for $i \neq j$, and maximal with this property: no further point of the sphere can be added. Maximality says every $u \in S^{d-1}$ is within ε of some u_i , so $\{u_i\}$ is an ε -net and $N \leq n$. Separation says the Euclidean balls $B(u_i, \varepsilon/2)$ are disjoint; they all lie in the ball $B(0, 1 + \varepsilon/2)$. Comparing volumes, $n(\varepsilon/2)^d \leq (1 + \varepsilon/2)^d$, which is the claim. ■

Conversely $N(S^{d-1}, \|\cdot\|, \frac{1}{4}) \geq 2^{d-2}$ (Exercise 2.8). A set of n points has $N \leq n$ at every scale; $[0, 1]$ with the metric $\sqrt{|t-s|}$ has $N \leq 1 + 1/(2\varepsilon^2)$ (Exercise 2.7).

2.3 Dudley's inequality

Theorem 2.4 (Dudley). *For every centered Gaussian process on (T, d) , with $D = \text{diam}(T)$,*

$$\mathbb{E} \sup_{t \in T} X_t \leq 12 \int_0^{D/2} \sqrt{\log N(T, d, \varepsilon)} d\varepsilon. \quad (3)$$

The integrand vanishes for $\varepsilon \geq D$, so the integral may be taken over $(0, \infty)$. For $T = \{1, \dots, n\}$ with $\text{Var } X_i \leq \sigma^2$: $D \leq 2\sigma$ and $N \leq n$, so (3) gives $\mathbb{E} \max_i X_i \leq 12\sigma\sqrt{\log n}$, Proposition 1.9 up to the constant.

Example 2.5 (The sphere). For $X_u = \langle g, u \rangle$ on S^{d-1} , $D = 2$ and Lemma 2.3 gives

$$\mathbb{E} \|g\| = \mathbb{E} \sup_{u \in S^{d-1}} \langle g, u \rangle \leq 12 \int_0^1 \sqrt{d \log(1 + 2/\varepsilon)} d\varepsilon = 12 c_0 \sqrt{d},$$

where $c_0 = \int_0^1 \sqrt{\log(1 + 2/\varepsilon)} d\varepsilon = 1.35$. The order $\sqrt{d}$ is sharp (Exercise 1.10); the constant 16.2 is not. The same computation on $S^{N-1} \times S^{d-1}$ gives $\mathbb{E} \|M\|_{\text{op}} \leq C\sqrt{N+d}$ for an $N \times d$ Gaussian matrix (Exercise 2.9).

2.4 Proof of Dudley's inequality: chaining

The union bound is applied to the *increments* of the process between successive nets. If $\pi_k(t)$ approximates t in an ε_k -net, $\varepsilon_k \downarrow 0$, then

$$X_t = X_{t_0} + \sum_{k \geq 1} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}),$$

the *chain*. Each link $X_{\pi_k(t)} - X_{\pi_{k-1}(t)}$ has standard deviation at most $3\varepsilon_k$, and there are at most $N(\varepsilon_k)^2$ distinct links at scale k , whatever the size of T . The union bound at scale k costs $3\varepsilon_k \sqrt{2 \log N(\varepsilon_k)^2}$; summing over k gives the integral. By (2) it suffices to bound $\mathbb{E} \max_{t \in F} X_t$ for finite $F \subset T$, using nets of T .

Proof of Theorem 2.4. If the integral is infinite there is nothing to prove, so assume $N(T, d, \varepsilon) < \infty$ for all $\varepsilon > 0$; then $D < \infty$. Fix a finite $F \subset T$.

Step 1: the nets. For $k \geq 0$ put $\varepsilon_k = D2^{-k}$ and let $T_k \subset T$ be an ε_k -net of T with $|T_k| = N_k := N(T, d, \varepsilon_k)$. Since $\varepsilon_0 = D$, we may take $T_0 = \{t_0\}$ a single point. For each $t \in T$ and $k \geq 0$ choose $\pi_k(t) \in T_k$ with $d(t, \pi_k(t)) \leq \varepsilon_k$. Then for every t and $k \geq 1$, by the triangle inequality,

$$d(\pi_k(t), \pi_{k-1}(t)) \leq d(\pi_k(t), t) + d(t, \pi_{k-1}(t)) \leq \varepsilon_k + \varepsilon_{k-1} = 3\varepsilon_k.$$

Step 2: the chain, and the union bound at each scale. For $K \geq 1$ and $t \in F$,

$$X_t - X_{t_0} = \sum_{k=1}^K (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}) + (X_t - X_{\pi_K(t)}),$$

so, taking the maximum over $t \in F$ term by term and then expectations (and $\mathbb{E}X_{t_0} = 0$),

$$\mathbb{E} \max_{t \in F} X_t \leq \sum_{k=1}^K \mathbb{E} \max_{t \in F} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}) + \mathbb{E} \max_{t \in F} (X_t - X_{\pi_K(t)}). \quad (4)$$

In the k th term the random variable $X_{\pi_k(t)} - X_{\pi_{k-1}(t)}$ depends on t only through the pair $(\pi_k(t), \pi_{k-1}(t)) \in T_k \times T_{k-1}$, so the maximum is over at most $N_k N_{k-1} \leq N_k^2$ centered Gaussians, each of variance $d(\pi_k(t), \pi_{k-1}(t))^2 \leq 9\varepsilon_k^2$ by Step 1. Proposition 1.9 gives

$$\mathbb{E} \max_{t \in F} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}) \leq 3\varepsilon_k \sqrt{2 \log N_k^2} = 6\varepsilon_k \sqrt{\log N_k}.$$

The last term in (4) is a maximum over $|F|$ centered Gaussians of variance at most ε_K^2 , hence at most $\varepsilon_K \sqrt{2 \log |F|}$, which tends to 0 as $K \rightarrow \infty$ because F is finite. Letting $K \rightarrow \infty$,

$$\mathbb{E} \max_{t \in F} X_t \leq 6 \sum_{k \geq 1} \varepsilon_k \sqrt{\log N(T, d, \varepsilon_k)}. \quad (5)$$

Step 3: the sum is an integral. Since $\varepsilon_k = 2(\varepsilon_k - \varepsilon_{k+1})$ and $N(T, d, \varepsilon) \geq N(T, d, \varepsilon_k)$ for $\varepsilon \leq \varepsilon_k$,

$$\varepsilon_k \sqrt{\log N_k} = 2 \int_{\varepsilon_{k+1}}^{\varepsilon_k} \sqrt{\log N_k} d\varepsilon \leq 2 \int_{\varepsilon_{k+1}}^{\varepsilon_k} \sqrt{\log N(T, d, \varepsilon)} d\varepsilon.$$

Summing over $k \geq 1$, the intervals $[\varepsilon_{k+1}, \varepsilon_k]$ tile $(0, \varepsilon_1] = (0, D/2]$, and (5) becomes $\mathbb{E} \max_{t \in F} X_t \leq 12 \int_0^{D/2} \sqrt{\log N(T, d, \varepsilon)} d\varepsilon$. The right side does not depend on F ; take the supremum over finite F . $\blacksquare$

Remark 2.6 (What the proof used). Only Proposition 1.9, once per scale, and the triangle inequality for d . The same proof applies to any process with $\mathbb{E}e^{\lambda(X_s - X_t)} \leq e^{\lambda^2 d(s,t)^2/2}$ (a *sub-Gaussian* process). The sum form (5) is often more convenient than the integral.

2.5 Sudakov's lower bound

The lower bound uses the covering number at a single scale.

Quoted: Sudakov's inequality. There is an absolute constant $c > 0$ such that for every centered Gaussian process on (T, d) and every $\varepsilon > 0$,

$$\mathbb{E} \sup_{t \in T} X_t \geq c\varepsilon \sqrt{\log N(T, d, \varepsilon)}.$$

Proof in Lecture 11, by comparison: on an ε -separated set of n points the increments of X dominate those of n independent Gaussians of variance $\varepsilon^2/2$; the Sudakov–Fernique inequality then compares the suprema, and Proposition 2.1 evaluates the independent one.

For the sphere, Exercise 2.8 gives $N(S^{d-1}, \|\cdot\|, \frac{1}{4}) \geq 2^{d-2}$, so Sudakov's inequality at $\varepsilon = \frac{1}{4}$ gives $\mathbb{E}\|g\| \geq \frac{\varepsilon}{4} \sqrt{(d-2) \log 2}$. Together with Example 2.5,

$$c' \sqrt{d} \leq \mathbb{E} \sup_{u \in S^{d-1}} \langle g, u \rangle \leq C \sqrt{d},$$

both from covering numbers. Neither bound is sharp for every T ; Talagrand's generic chaining replaces the integral by a quantity that is, and is not needed here.

Exercise 2.7 (TYPE A). Two processes on $[0, 1]$. (a) Brownian motion has canonical metric $d(s, t) = \sqrt{|t - s|}$. Show $N([0, 1], d, \varepsilon) \leq 1 + \frac{1}{2\varepsilon^2}$ and conclude from Theorem 2.4 that $\mathbb{E} \sup_{t \leq 1} B_t < \infty$: the maximum over a continuum of times is a random variable with finite mean. *The bound is 9.7; the truth is $\sqrt{2/\pi} \approx 0.80$.* (b) Let X be a process on $[0, 1]$ with $\text{Var } X_t \leq 1$ and $d(s, t) \leq L|t - s|$ for some $L \geq 1$: it changes on the scale $1/L$. Show $N([0, 1], d, \varepsilon) \leq 1 + \frac{L}{2\varepsilon}$ and conclude

$$\mathbb{E} \sup_{t \in [0, 1]} X_t \leq 12\sqrt{\log(1 + L)} + 12.$$

You may use $\int_0^1 \sqrt{\log(A/\varepsilon)} d\varepsilon \leq \sqrt{\log A} + 1$ for $A \geq 1$. Compare with $\sqrt{2 \log L}$, the maximum of L independent coordinates. (c) The union bound on the grid $\{j/n\}$ gives $\mathbb{E} \max_j B_{j/n} \leq \sqrt{2 \log n}$, which is useless as $n \rightarrow \infty$, while (a) is uniform in n . In one sentence: what does chaining use that the union bound throws away?

Exercise 2.8 (TYPE B). The sphere from below. (a) Let $u_1, \dots, u_n$ be an ε -net of S^{d-1} with $\varepsilon \leq 1$. Show that every x with $1 - \varepsilon \leq \|x\| \leq 1$ lies within 2ε of some u_i , and compare volumes to get $n(2\varepsilon)^d \geq 1 - (1 - \varepsilon)^d$. (b) Deduce $N(S^{d-1}, \|\cdot\|, \frac{1}{4}) \geq 2^{d-2}$ for $d \geq 2$. (c) Conclude from Sudakov's inequality that $\mathbb{E}\|g\| \geq c' \sqrt{d}$ for an absolute constant c' , and compare with Example 2.5: both sides of $\mathbb{E}\|g\| \asymp \sqrt{d}$ come from counting balls.

Exercise 2.9 (TYPE B). The operator norm by chaining. Let M be $N \times d$ with i.i.d. $\mathcal{N}(0, 1)$ entries and $Y_{(u,v)} = \langle v, Mu \rangle$ on $T = S^{N-1} \times S^{d-1}$, as in Example 1.5, so that $\sup_T Y = \|M\|_{\text{op}}$. (a) Show that the canonical metric is $d((u, v), (u', v')) = \|vu^\top - v'(u')^\top\|_F$, and that it is at most $\|u - u'\| + \|v - v'\|$. (b) Show $N(T, d, \varepsilon) \leq (1 + 4/\varepsilon)^{N+d}$. (c) Conclude $\mathbb{E}\|M\|_{\text{op}} \leq C\sqrt{N+d} \leq C(\sqrt{N} + \sqrt{d})$ for an absolute constant C . *Dudley gives $C = 19$; for $N = 100$, $d = 50$ the truth is 16.6 and $\sqrt{N} + \sqrt{d} = 17.1$. Lecture 11 gets the constant 1 by comparison.*

ASSIGNED Lecture 10, due at the start of Lecture 11: Exercises 2.7, 2.8, 2.9. *Three problems: Dudley on two processes on $[0, 1]$; the sphere from below; the operator norm with the wrong constant.*

Lecture 11 compares two Gaussian processes by interpolating between them, proves Sudakov's inequality, and reads the operator norm of a Gaussian matrix off the comparison.